

MODULE 10

SQL Indexes and Performance

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL Indexes and Performance

What is an Index?

Speeds up data retrieval dramatically.

Without index: full table scan (slow).

With index: direct access (fast).

Trade-off: faster reads, slower writes.

Index Types

PRIMARY KEY: automatic clustered index

UNIQUE INDEX: enforces uniqueness + fast lookup

REGULAR INDEX: speeds up queries

COMPOSITE: index on multiple columns

FULLTEXT INDEX: for text search

Creating Indexes

```
CREATE INDEX idx_name ON students(last_name);  
CREATE UNIQUE INDEX idx_email ON students(email);  
CREATE INDEX idx_dg ON students(department, gpa);  
DROP INDEX idx_name ON students;
```

EXPLAIN

EXPLAIN SELECT * FROM students WHERE email = 'a@b.com';

type = ref/const --> index used (good)

type = ALL --> full scan (bad, add index!)

key --> which index was used

rows --> estimated rows scanned

Best Practices

Index WHERE, JOIN ON, ORDER BY columns

Index high-cardinality columns

Use composite indexes for common query patterns

Do NOT index every column

Do NOT index rarely-used columns